

Does Increasing Women's Schooling Raise the Schooling of the Next Generation? Reply

By JERE R. BEHRMAN AND MARK R. ROSENZWEIG*

Kate L. Antonovics and Arthur S. Goldberger (henceforth AG) raise questions about the conclusions in the article by Jere R. Behrman and Mark R. Rosenzweig (2002), henceforth BR. In particular, they are unsure about how the BR twins-based estimates inform policy, and they seek to show that the conclusions of the study are not robust, based on alternative codings of the schooling data and by adding observations. In our response, we first discuss the relationship between the empirical findings in BR and policy questions. We show that the focus of AG's comments is misguided, given the policy issues. We then reassess the empirical robustness of the BR results using new information on schooling which was collected and coded independently of both AG and BR. We conclude that the independently collected and coded data, and even the new codings by AG, provide additional support for the original BR results showing that the positive cross-sectional relationship between a mother's schooling and her child's schooling is not robust to controls for unmeasured, intergenerationally correlated endowments, while the positive effect of paternal schooling on children's schooling is robust. One justification for policies, now in place in a number of countries, designed to raise women's schooling relative to men's thus does not appear to have empirical support.

I. Are the BR Estimates Policy-Relevant?

In BR we consider a linear reduced-form equation determining the schooling of child i in family j , which we now write as:

$$(1) \quad S_{ij}^c = \delta_1 S_j + \delta_2 S_j^s + \Gamma_1 r e_j + \Gamma_2 r e_j^s + \gamma r + r \varepsilon_{ij}^c$$

where the subscripts i and j indicate that the dependent variable is for the i th individual in family j and the superscript c denotes that the individual is a child, S_j is the schooling of the child's mother, S_j^s is the schooling of the child's father, the e 's are the preschool endowments of the two parents, r is the return to skill, and ε_{ij}^c is a child-specific endowment. Equation (1) differs from the schooling equation in BR because it decomposes the endowments into skill and return components and adds a parameter γ incorporating the fact that school investment may respond to changes in the skill return r . We show in the paper that it is possible to identify the parameters δ_1 , δ_2 , Γ_1 , and Γ_2 using cross-sectional information from samples of monozygotic (MZ) twin mother and father pairs even though the endowments are not directly measurable.¹ AG raise the question of whether identification of these parameters is useful for policy purposes, as we claim, and note that we did not provide one "concrete" example of "an intervention that increases the overall level of women's schooling." It is true that we did not explicitly link the parameters in (1) to specific policy interventions; nor did we cite any actual policy focusing on raising the schooling of women. We now do both and show, as well, that the estimates that AG focus on, presented in their Table 1, are not informative compared with the estimates in their Table 3 for assessing actual policy interventions.

Initiatives designed to raise women's schooling relative to men's schooling are high on the agenda of policymakers in many developing

* Behrman: Department of Economics, University of Pennsylvania, 160 McNeil Building, Philadelphia, PA 19104 (e-mail: jbehrman@econ.upenn.edu); Rosenzweig: Department of Economics, Yale University, 27 Hillhouse Avenue, New Haven, CT 06520 (e-mail: Mark.Rosenzweig@yale.edu). This research was supported in part with funds from NIH Grant RO1 HD046144-01, NIA Grant RO1 AG11725-01A1, and NSF Grant SBR95-11955.

¹ It is possible that r varies across time and thus across cohorts. The twins in the sample were making schooling decisions for their children in the mid-1960s through the mid-1980s. Sample size precludes estimating precise schooling effects by cohort.

countries and at international organizations such as the World Bank and the United Nations,² in large part predicated on the notion that these would be efficacious in raising human capital investments in children.³ And interventions have been undertaken to accomplish this goal in a number of countries. Programs exist in Pakistan, for example, which provide fellowships to families exclusively for the schooling of girls.⁴ The PROGRESA Program in Mexico provides subsidies for school attendance that are higher for girls than for boys.⁵ An important national program introduced by the government of Bangladesh in the early 1990s provides both tuition subsidies and stipends to families, conditional on secondary school attendance by girls but not boys. This program now covers all of the rural population of Bangladesh. A recent careful analysis of the program based on household survey data (Shahidur Khandker et al., 2003) indicates that these interventions raised the schooling of the girls but not of the boys.

The Bangladesh policy intervention is an example of a program that raises the schooling of mothers, but not fathers. What is the effect on the schooling of the next generation, purported to be a benefit of such policies? First, in general equilibrium, the more educated women cannot have better-educated husbands if all women marry, so there is a minimal assortative mating effect. However, such an intervention may affect the returns to skill r . The general-equilibrium, intergenerational schooling effect of an intervention that raises the schooling of women only is thus

$$(2) \quad dS_{ij}^c/dS_j = \delta_1 + [\Gamma_1 e_j dr/dS_j + \Gamma_2 e_j^c dr/dS_j + \gamma dr/dS_j + \varepsilon_{ij}^c dr/dS_j].$$

It is reasonable to assume that raising the schooling level of women will lower the return to skills in equilibrium. Our estimates of the parental endowment effects Γ 's are positive, and it is also reasonable to assume that γ is positive (a higher r induces more investment in schooling). Thus, the bracketed term in (2) is likely to be negative and the twin-based estimate of δ_1 is likely an overestimate of the effect of the policy intervention. We find that the effect of increasing maternal schooling on child schooling conditional on own and husband's endowments δ_1 is not positive, so the general-equilibrium policy effect is likely not positive. Moreover, if the general-equilibrium effects on returns to skills r of raising women's or men's schooling are similar, then the difference between δ_1 and δ_2 indicate whether policy interventions favoring boys or girls would be preferred if the criterion is the impact on the schooling of the next generation, assuming the costs of the interventions are the same. We find that δ_2 is positive, suggesting that if all countries were like the United States, interventions focusing on women's schooling in order to advance the schooling of the next generation would be misguided.⁶

AG's final point is that any intervention might alter not only the returns to S_j but also the ε_{ij}^c 's directly to the extent that they reflect pre-school investments and not just immutable genetics. We assume in (2) (for analyzing policy, not for econometric identification of δ_1), that the ε_{ij}^c 's do not change. However, we find that δ_1 , the response of schooling investment to changes in maternal schooling, is small. It is necessary that preschool investments are more responsive than schooling investments to maternal schooling for the estimate of δ_1 , based on twin pairs, to be a significantly downward biased estimate of the policy response. In principal, one could use the same twins-based methodol-

² Indeed the Millenium Development Goals established in 2000, which are increasingly shaping international debates about policies in developing countries, explicitly include increasing schooling for females relative to males.

³ There are a number of academic papers that suggest the same possibility. For example, Heckman and Hotz (1986) state, "Such evidence would seem to indicate the importance of the mother in providing a learning environment for her child. It demonstrates a potentially important nonmarket benefit of mother's education that should be evaluated in assessing the value of subsidies to women's education" (Heckman and Hotz, 1986, p. 532).

⁴ For descriptions of these programs, see Jooseop Kim et al. (1999).

⁵ For description and analysis of this program, see Behrman et al. (2005) and Paul T. Schultz (2004).

⁶ BR also show that increasing the schooling of women increases their participation in the labor market. This enhances the negative general-equilibrium effect on skill returns. However, the increased participation by women may directly affect the demand for schooling by girls by altering norms of behavior and/or through "role-model" effects.

ogy to examine to what extent mother's and father's schooling affect preschool investments, controlling for endowments. AG present no such evidence.

Despite their concern about policy-relevance, AG focus on the effects of an increase in a person's schooling on his or her children's schooling, *inclusive* of the assortative mating effect, rather than on the parameters δ_1 and δ_2 , in their Table 1. No rationale is given as to why these are of interest. They present estimates of

$$(3) \quad dS_{ij}^c/dS_j = \delta_1 + \delta_2 dS_j^s/dS_j + \Gamma_2 rde_j^s/dS_j$$

$$\text{and } dS_{ij}^c/dS_j^s = \delta_2 + \delta_1 dS_j/dS_j^s + \Gamma_1 rde_j/dS_j^s$$

and compare the effects for each parent. The difference across the parents between these unconditional intergenerational schooling effects, ignoring for simplicity the sorting by unmeasured endowments, is

$$(4) \quad dS_{ij}^c/dS_j - dS_{ij}^c/dS_j^s = \delta_1(1 - dS_j/dS_j^s) \\ - \delta_2(1 - dS_j^s/dS_j),$$

which can be very small if there is a high degree of assortative mating (dS_j^s/dS_j or dS_j/dS_j^s close to 1), even if the conditional schooling effects δ_1 and δ_2 are very different for mothers and fathers. Given our finding that dS_j^s/dS_j is 0.38, AG's conclusion that for some measurement variants (4) is small is thus uninformative for assessing gender-specific policy interventions. We thus focus our discussion on the robustness of the estimates of approximations to the individual parameters δ_1 and δ_2 that AG present in Table 3.

II. Robustness and the Measurement of Parental Schooling

A major issue in obtaining good estimates of schooling effects using within-sibling methods is measurement error. In designing the questionnaire for our survey of twins, BR and the late Paul Taubman (BRT), with the advice of the Temple University Survey Research Center, composed a set of 20 questions about schooling

accomplishments in order to construct more accurate schooling measures. Constructing singular measures of schooling from these answers proved to be complex, and a number of cases required judgments in going from the 20 responses to a single number. Thus, it is useful to have had a second set of researchers go through the set of answers to construct their own schooling measures, as AG have done, to assess robustness. Of course, BRT also elicited cross-twin reports on schooling and BR used these in their models to deal with the inevitable measurement errors, as in prior twin-based studies of earnings determinants, procedures eschewed by AG.

AG's estimates in their Table 3 using their own coding schemes and sample choices do not contradict any of BR's main conclusions. For 18 of the 20 samples and variants of parental schooling measures, they find that the within-MZ twins estimates of maternal schooling effects are lower than those for fathers, which are always positive. And in the sample of female twins, which we argue in BR provide the less biased estimates of maternal schooling effects, in two of three of the cases, excluding observations that BR did not use in their analysis, the maternal schooling effect is negative using the AG coding. These results are not surprising. As shown in their Table 2, ignoring AG's nonstandard coding, which added in years of vocational schooling, it is possible that in only 7 percent of the pairs did their alternative schooling measurement yield nonidentical, within-pair schooling differences.⁷

The reader might seek a basis for choosing between the AG and BR coding schemes on the basis of a measure of schooling that is independent of both AG's and BR's coding. Fortunately, we have a third set of independently acquired and coded school measurements, provided by the Minnesota Twin Registry (MTR). In the 1980s, the MTR mailed a biographical questionnaire to the twins which contained two items eliciting own schooling:

⁷ AG did not document how they allocated coding discrepancies by type in that unknown number of cases in which there were more than two discrepancies, so it is not possible to know the exact proportion.

TABLE 1—COMPARISONS OF BR AND AG CODING OF TWINS SCHOOLING WITH BQ CODING

Sample coding	Female MZ pairs		Male MZ pairs	
	BR	AG	BR	AG
Observations above	72	114	34	79
Observations match	232	263	136	189
Observations below	59	56	53	35

“Circle highest grade completed:

9 10 11 12”

“Circle years of college:

0 1 2 3 4 5 6 7 8 9.”

The twins were also asked to provide their highest degree. We have the MTR-coded years of own schooling for all of the twins who answered these questions. These make up 85 percent of the twins in the subsamples used by BR and AG.

How do the AG and BR own measures of schooling based on the 1994 survey compare to the earlier MTR measures? The correlations between the AG and the BR measures of own schooling with the MTR self-reports are 0.89 and 0.88. However, the AG measures appear to be biased upward relative to the MTR measures. Table 1 reports the number of observations that match, are above, or are below the original MTR schooling-years measures for both of the 1994 survey-based measures in the fully overlapping cases. For maternal schooling, the BR measures match the original measures in 4 percent more cases than do the AG measures. Moreover, the deviations for the BR measures are symmetric, consistent with classical measurement error models, but those of AG are biased positively, with more than twice as many observations above as below the MTR schooling measures.^{8,9} The same bias is seen for the

AG measures for fathers, although there are slightly more exact matches for the AG than for the BR measures.

It is certainly not the case that these statistics indicate that the AG coding, and the findings based on them, should be preferred to those of BR. As an additional assessment of the robustness of the BR findings to coding interpretations of maternal schooling, however, we can use the MTR schooling measures in place of the BR and AG measures for respondent schooling. Because in 41 percent of the cases the twins did not report the highest degree or grade for the spouse in the MTR questionnaire, sample size precludes obtaining estimates of maternal schooling effects from the male MZ sample. Moreover, for the same reason, it is necessary to use the 1994 questionnaire information for the husband’s schooling in the female MZ sample. Table 2 reports for the female MZ twins the estimates for some of the same specifications AG used for Table 3, but uses the MTR measures of maternal schooling. The variants are based on whether the father’s schooling uses the BR or the AG coding, whether the extra observations used by AG but not by BR are included, and whether children under 18 are excluded. The AG coding for children’s schooling is used in all cases. As can be seen, the within-estimate of the intergenerational maternal schooling effect δ_1 is negative, while the paternal schooling effect δ_2 is positive in every variant, just as in the original BR study. Of course, these individual estimates of δ_1 and δ_2 are inferior to those of BR because they take into account neither measurement error (which is evidently nontrivial) nor variation in spouse endowments. The key points are that the evidence suggests that the qualitative conclusions of the BR study are robust to independent codings of schooling and that the AG codings are obviously not the preferred ones.

⁸ This is of course mostly due to AG’s including vocational years of schooling, contrary to conventional practice.

⁹ For a younger sample one might expect that adults returning to school between the two measurements could result in an asymmetric upward pattern such as indicated for the AG estimates. However, the twins in the sample were

from 39 to 58 at the time of the BRT survey in 1994, so for the birth cohorts covered, returning to school as adults is not likely to account for the AG patterns in Table 1. Moreover, for these cohorts, females are thought to have returned to school more than males because their schooling was interrupted by birth and care of small children. But the asymmetry for the AB coding is equally important for men as for women.

TABLE 2—SLOPES IN MULTIPLE REGRESSIONS OF CHILD'S SCHOOLING ON BOTH PARENTS' SCHOOLING, USING BQ INFORMATION ON MOTHER'S SCHOOLING: FEMALE MZ TWIN PAIRS

Database	Coding of father's schooling	N	Parent	Cross	Within
BROAD	BR	168	Mother	0.117 (0.073)	-0.211 (0.169)
			Father	0.266 (0.059)	0.125 (0.079)
	BR/WS	167	Mother	0.101 (0.074)	-0.191 (0.167)
			Father	0.293 (0.059)	0.175 (0.081)
	WS	200	Mother	0.214 (0.074)	-0.054 (0.153)
			Father	0.251 (0.057)	0.122 (0.074)
NARROW	BR	132	Mother	0.138 (0.079)	-0.135 (0.199)
			Father	0.228 (0.064)	0.139 (0.095)
	BR/WS	131	Mother	0.111 (0.083)	-0.124 (0.196)
			Father	0.262 (0.068)	0.199 (0.099)
	WS	156	Mother	0.224 (0.081)	-0.078 (0.179)
			Father	0.208 (0.061)	0.123 (0.086)

Notes: BROAD = children 10 and over. NARROW = children aged 18 and over. N = number of twin pairs. Standard errors in parentheses, corrected for clustering in pairs for the cross-sectional estimates.

III. Robustness and Choice of Observations

The AG results in their Table 3 that are least consistent with those of BR are those based on a sample containing additional observations not used by BR. The augmented-sample within-estimates still indicate that paternal exceeds maternal schooling effects, but both effects are insignificant and the maternal effect is not negative. The supplemental observations, however, are suspect. Observations were dropped by BR for two reasons. If any twin in any pair was missing any of this spousal information, the twin pair was excluded from the BR analysis. In addition, if there were unresolvable ambiguities about children's schooling, observations were excluded. The first criterion was used because, to obtain the policy-relevant estimates of the parameters in

equation (1), information on the work experience and earnings of the spouses, in addition to information on the schooling of the parents, was required. Because AG were not interested in obtaining estimates of the policy parameters, and required only schooling information, they could employ a larger sample. However, they also evidently included cases for which information on children's schooling was ambiguous. They were able to add 49 additional female and 51 male twin pairs using less stringent criteria than BR. We could find all but two of the female twin pairs.

There are a number of reasons to be suspicious of the added observations based on their observed characteristics. In this additional subset, 21 percent of the twins were in their second or higher-order marriage, compared to only 13 percent in the original sample. Multiple marriages lead to ambiguities as to which spouse is the relevant father, particularly in the female sample. More disturbingly, 11 percent of the children have schooling or expected schooling measured to be below 12 years. This compares with only 2.4 percent in the original sample, and is also far below schooling attainment figures for the population of Minnesota. Closer inspection of the data reveals that 57 percent of the children with below 12 years of schooling in this subsample were 18 years of age or less, even though only a third of all of the subsample children were below age 18. Unless there is an unnoticed secular trend to lower schooling in Minnesota, it is likely that incomplete schooling has been coded as complete. Indeed, in the original sample, only 19 percent of those children with less than a high school education were under 18 years of age, compared to 29 percent overall. In fact, we find that in 16 twin pairs, one child was reported as still in school and was below age 18, but there was no parental information on schooling expectations. These children were coded with their so-far completed schooling as their lifetime schooling. These cases are clearly invalid. In 14 more twin pairs, at least one child was at the grade level consistent with his or her age (e.g., age 14 and completed the eighth grade) but was reported by the parents to be not in school. AG coded these cases using the completed grade level (8 in this example). These are also ambiguous cases, given summer recess, and the choices made are not likely to be correct. None of these pairs was included in the original BR sample.

TABLE 3—SLOPES IN CROSS-SECTION (OLS) REGRESSIONS OF CHILD'S SCHOOLING ON BOTH PARENTS' SCHOOLING, BY SAMPLE, USING AG CODING

Sample	Female MZ pairs		Male MZ pairs	
	Overlap original	Added	Overlap original	Added
BROAD				
Number of twin pairs	202	47	120	51
Mother	0.131 (0.066)	0.738 (0.162)	0.330 (0.082)	0.055 (0.144)
Father	0.283 (0.053)	-0.020 (0.147)	0.271 (0.070)	-0.082 (0.143)
NARROW				
Number of twin pairs	154	28	76	27
Mother	0.164 (0.075)	0.625 (0.239)	0.250 (0.102)	0.051 (0.179)
Father	0.243 (0.058)	0.066 (0.168)	0.260 (0.086)	0.087 (0.55)

Note: Standard errors in parentheses, corrected for clustering in pairs.

Neither the descriptive statistics on the added observations nor the fact that the cases lacked important information are sufficient to rule them out completely. However, the simple cross-sectional relationships between measured parental schooling and child schooling in these sets of observations are strikingly different from the cross-sectional findings in the original BR data and in the prior literature. Table 3 compares the cross-sectional estimates of the schooling effects for the BR/AG overlap sample of twin pairs and for the observations added by AG for both the "broad" and "narrow" age definitions of children.¹⁰ We use the AG coding for all schooling and child age measures. All four overlap cross-sectional sample estimates conform to the cross-sectional findings from previous studies of intergenerational schooling effects based on United States data, such as Behrman and Paul Taubman (1985, 1986). These studies, and the overlap data from all four subsamples, indicate that the schooling of both parents is statistically significantly and positively related to the schooling of the children. The added sample estimates are quite different. The estimates of the paternal schooling effect in all of the added subsamples are statistically

insignificant and substantially smaller in magnitude than those in the corresponding overlap samples, and are actually negative in both the female and male broad subsamples. And neither paternal nor maternal schooling is significantly related to child schooling in the male sample of twin pairs for either age group of children: the point estimates for female schooling are from 80 percent (narrow sample) to 83 percent (broad sample) lower than those estimated from the overlap sample using BR criteria. It is not surprising, therefore, that the BR findings are not fully robust to the addition of these strange outliers, which were excluded originally based on missing or ambiguous information.

The conclusions of BR were that the strong positive cross-sectional partial correlation between a mother's schooling and her child's schooling is not robust to the elimination of endowment effects, while the estimated positive effect of the husband's schooling is robust. These conclusions now appear to be themselves robust to additional measurement errors in schooling coding, if not fully to the use of observations with ambiguous information, barring reasonably precise measurements of children's schooling or the identification of the appropriate parents.

REFERENCES

- Behrman, Jere R. and Rosenzweig, Mark R.
 "Does Increasing Women's Schooling Raise the Schooling of the Next Generation?"

¹⁰ There were also observations used in the original BR samples that were excluded by AG. However, these are too small in number to analyze separately: 10 pairs and 2 pairs in the female and male samples, respectively. All standard errors are corrected for pair clustering, as in the original BR cross-sectional estimates.

- American Economic Review*, 2002, 92(1), pp. 323–34.
- Behrman, Jere R.; Sengupta, Piyali and Todd, Petra.** “Progressing through PROGRESA: An Impact Assessment of Mexico’s School Subsidy Experiment.” *Economic Development and Cultural Change* (2005).
- Behrman, Jere and Taubman, Paul.** “Intergenerational Earnings Mobility in the United States: Some Estimates and a Test of Becker’s Intergenerational Endowments Model.” *Review of Economics and Statistics*, 1985, 67(1), pp. 144–51.
- Behrman, Jere R. and Taubman, Paul.** “Birth Order, Schooling, and Earnings.” *Journal of Labor Economics*, 1986, 4(3), pp. S121–45.
- Heckman, James J. and Hotz, V. Joseph.** “Investigation of Labor Market Earnings of Panamanian Males: Evaluating the Sources of Inequality.” *Journal of Human Resources*, 1986, 21(4), pp. 507–42.
- Khandker, Shahidur; Pitt, Mark and Fuwa, Nobuhiko.** “Subsidy to Promote Girls Secondary Education: The Female Stipend Program in Bangladesh.” 68th Annual Meeting of the Population Association of America. Minneapolis, MN, May 1–3, 2003.
- Kim, Jooseop; Alderman, Harold and Orazem, Peter F.** “Can Private School Subsidies Increase Enrollment for the Poor? The Quetta Urban Fellowship Program.” *World Bank Economic Review*, 1999, 13(3), pp. 443–65.
- Schultz, T. Paul.** “School Subsidies for the Poor: Evaluating the Mexican Progreso Poverty Program.” *Journal of Development Economics*, 2004, 74(1), pp. 199–250.